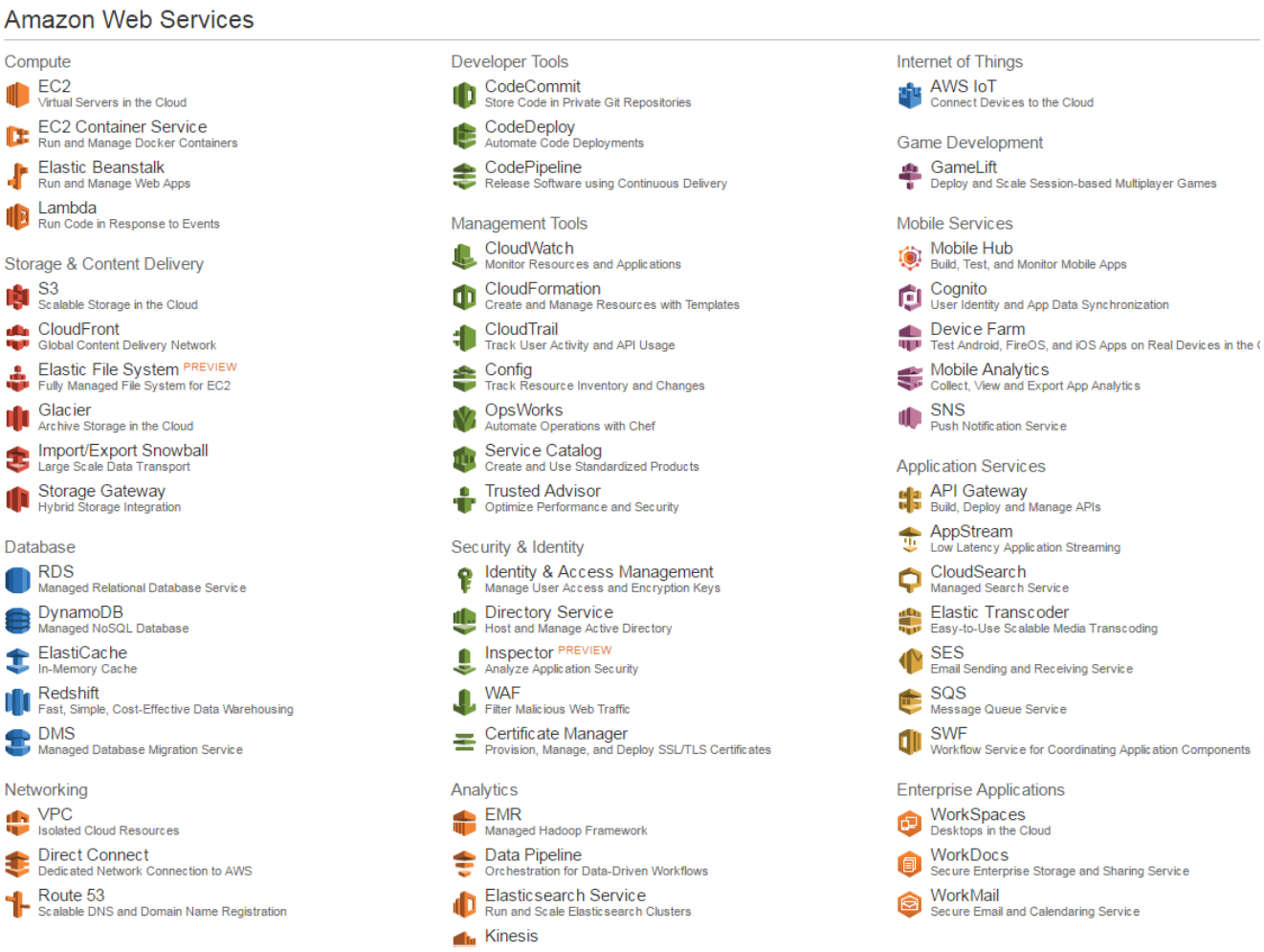
AWS Auto Scaling Configuration by kvreddi

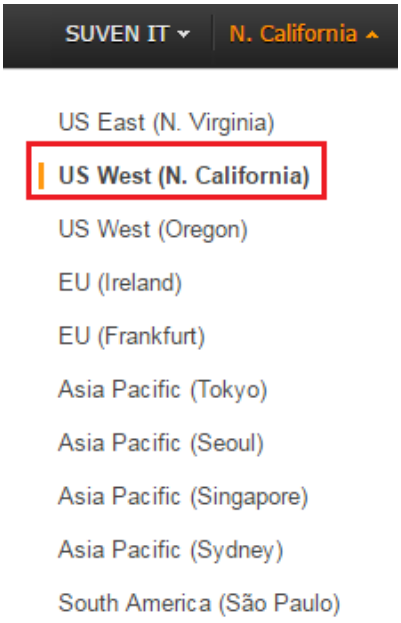
STEP#1: Login to Amazon Web Service Console



The AWS Management Console is a web control panel for managing all your AWS resources, from EC2 instances. The Console enables cloud management for all aspects of the AWS account, including managing security credentials, or even setting up new IAM Users.

STEP#2: Select the right AWS Region

Amazon Web Services is available in different Regions all over the world and the Console lets you provision resources across multiple regions. You usually choose a region those best suits your business needs to optimize your customer's experience

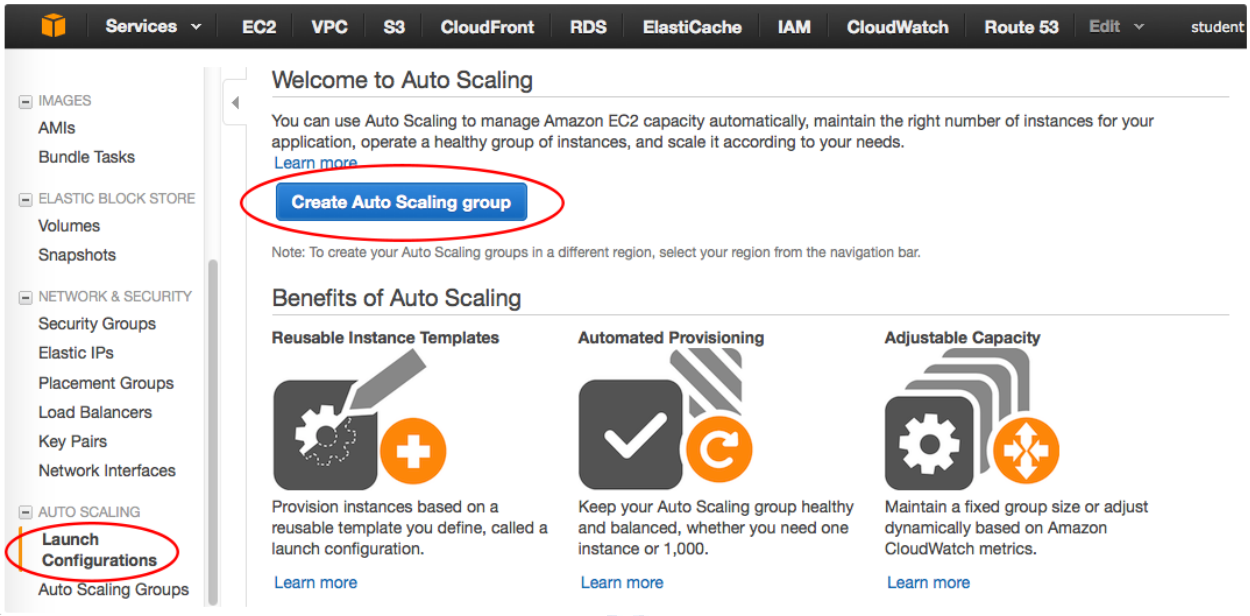


Create an Auto Scaling Configuration

A **Launch Configuration** is a template that the Auto Scaling group uses to launch Amazon EC2 instances. If you've launched an individual EC2 instance before, you've already walked through the process of defining compute characteristics such as the instance type, security groups, and configuration scripts. A launch configuration allows you to define these same characteristics, which are then applied to any instances launched into the Auto Scaling group.

You create the launch configuration by including information such as the Amazon machine image ID to use for launching the EC2 instance, the instance type, key pairs, security groups, and block device mappings, among other configuration settings. When you create your Auto Scaling group, you must associate it with a launch configuration. You can attach only one launch configuration to an Auto Scaling group at a time and it cannot be modified.

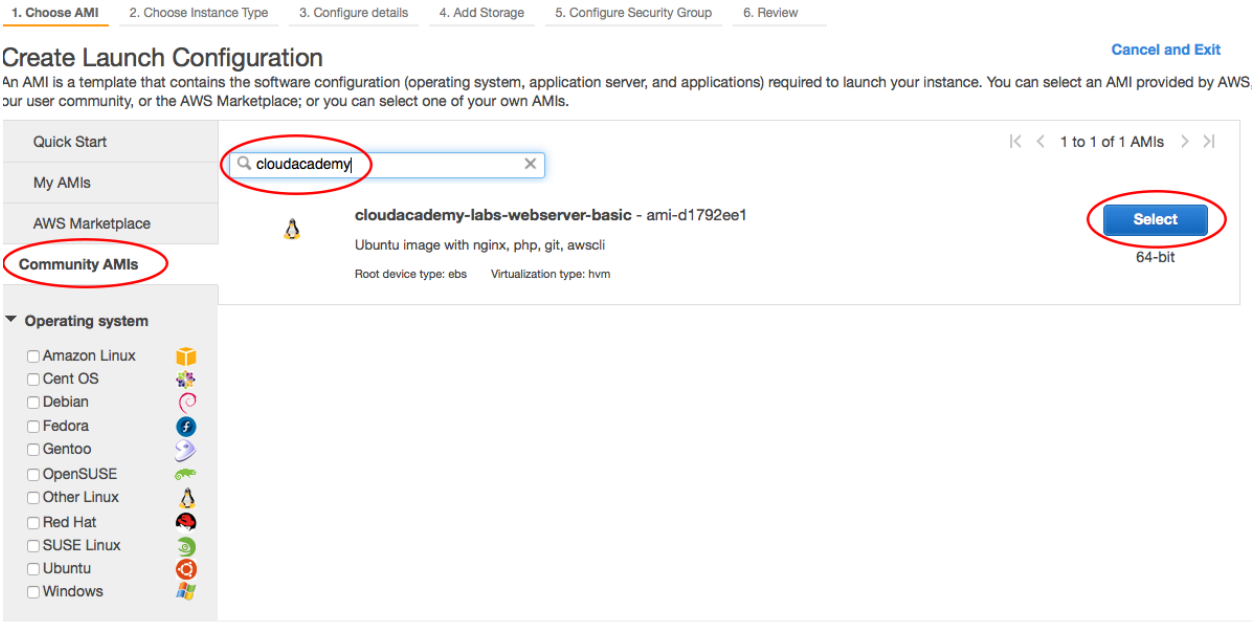
To start creating an Autoscaling Group, open the **Launch Configurations** page and click on the **Create Auto Scaling group**button.



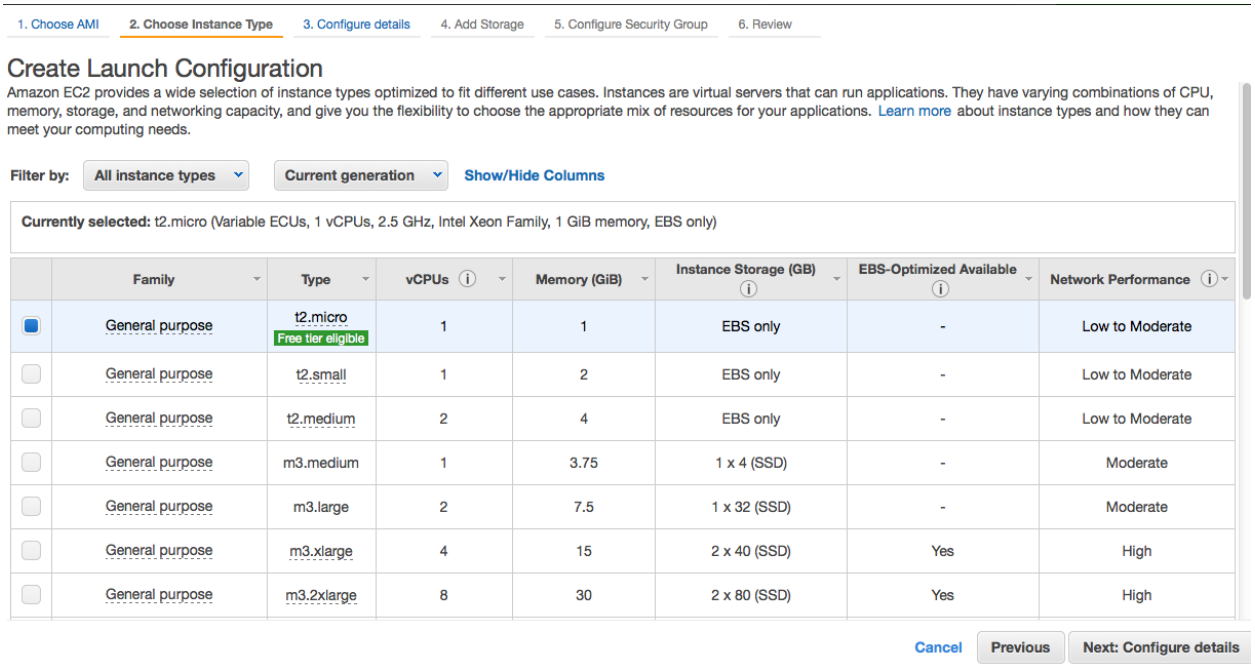
Then click on **Create Launch configuration**. By there AWS Management Console guides you though each required step and displays to you a graphical interface that is quite similar to the Launch Instance Wizard one.

The first step to accomplish is the **AMI selection;** you have to select the AMI that will be used by all the EC2 instances of the Auto Scaling group. The Cloud Academy DevOps team created a specific AMI for this laboratory, you can find it by searching the word "cloudacademy" in the AMI search bar.

Select "**cloudacademy-labs-webserver-basic**" AMI and click on Select.

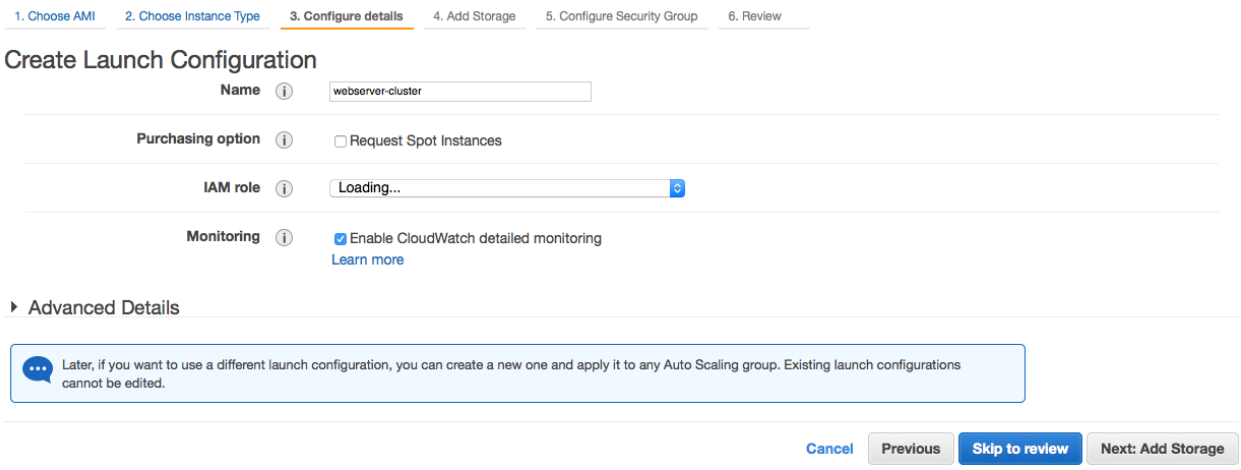


The next step is choosing the instance type: select the t2.micro type and click on **Next, Configure details** button.



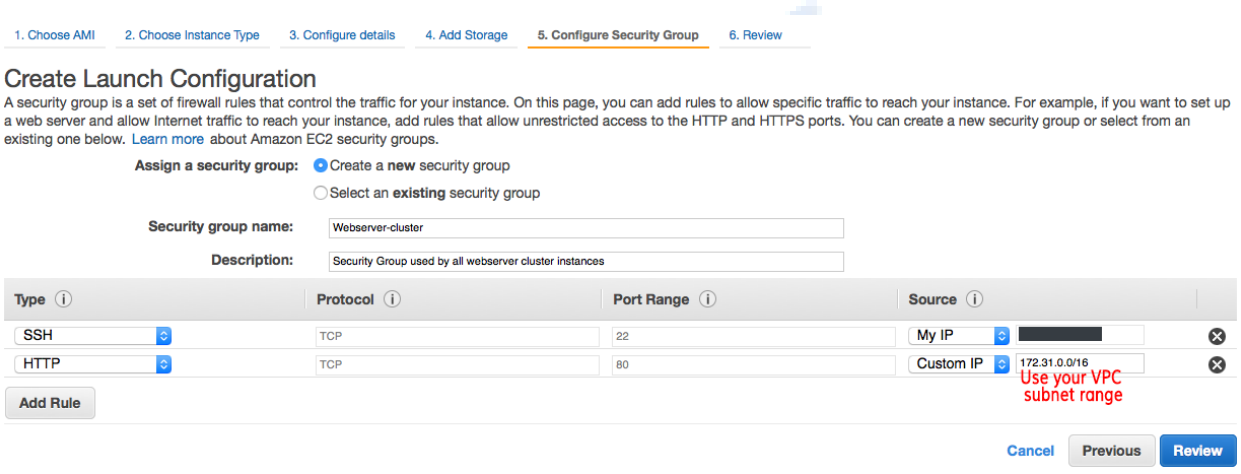
The **Configure details** step asks to you a name for your launch configuration and if you want to enable the CloudWatch monitoring for your future instances.

Click on **Next: Add Storage** when you filled the required fields.

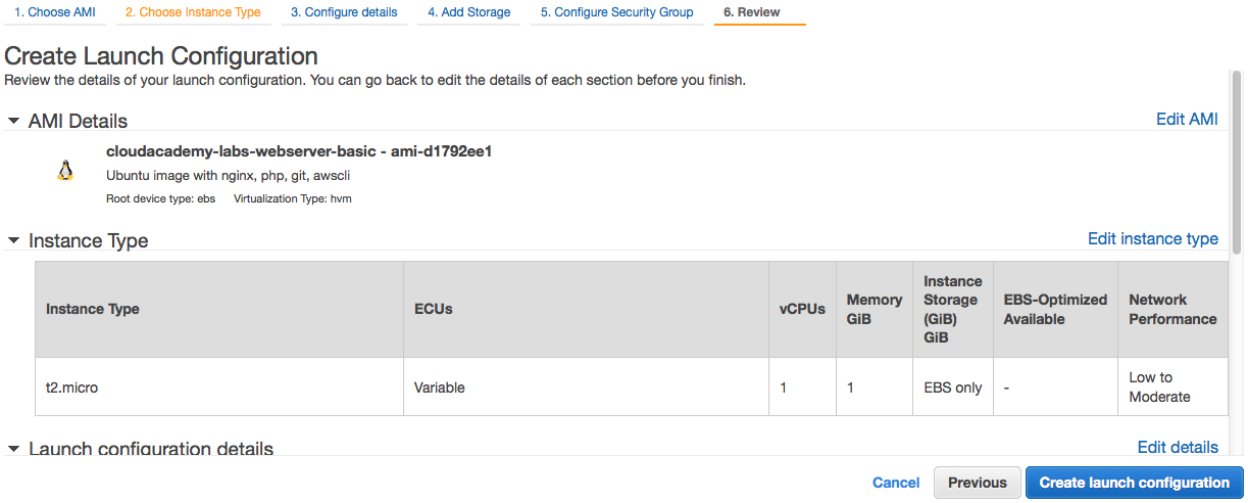


Create a new Security Group for your Auto Scaling Group, choose a name and a description and add the required rules to use SSH and HTTP.

The default Amazon VPC subnet range is `172.31.0.0/16`. You can use it to allow the HTTP traffic, so the Elastic Load Balancing instance will be able route the HTTP requests to the instances of the Auto Scaling group.



review the Launch Configuration and create the Keypair that will be injected in any instance created by the AS service.



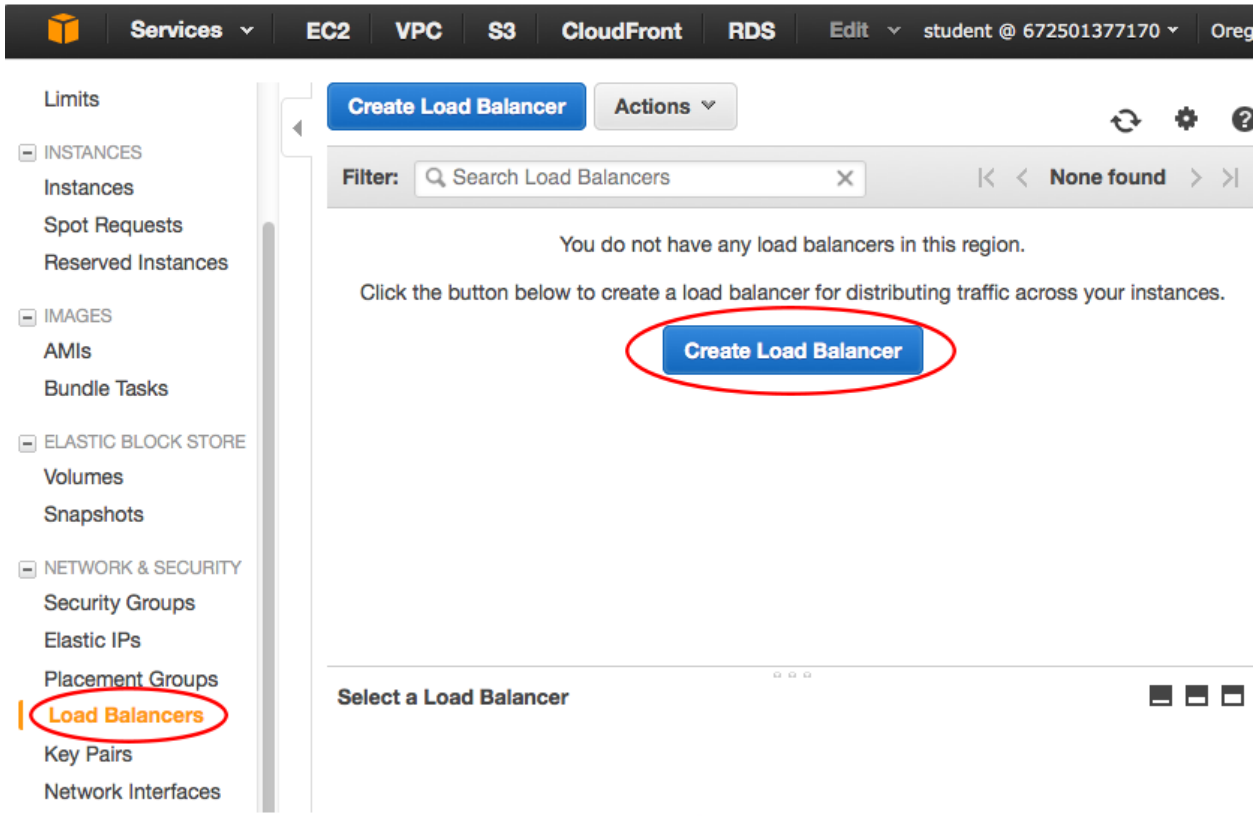
Click on **Create launch configuration** and you are ready to create your first Auto Scaling Group.

Create a load balancer using ELB

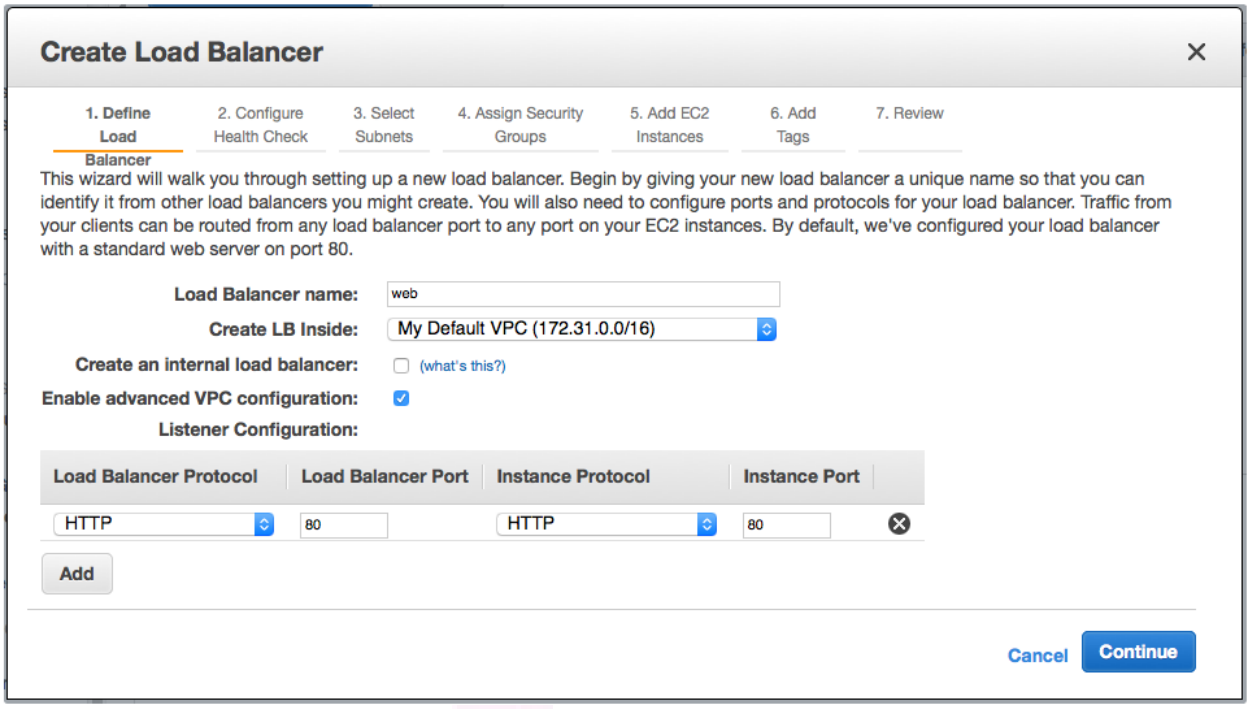
Elastic Load Balancing automatically distributes incoming application traffic across multiple Amazon EC2 instances. It enables you to achieve even greater fault tolerance in your applications, seamlessly providing the amount of load balancing capacity needed in response to incoming application traffic. Elastic Load Balancing detects unhealthy instances within a pool and automatically reroutes traffic to healthy instances until the unhealthy instances have been restored. Customers can enable Elastic Load Balancing within a single Availability Zone or across multiple zones for even more consistent application performance.

You can create your first Load Balancer by taking the following steps:

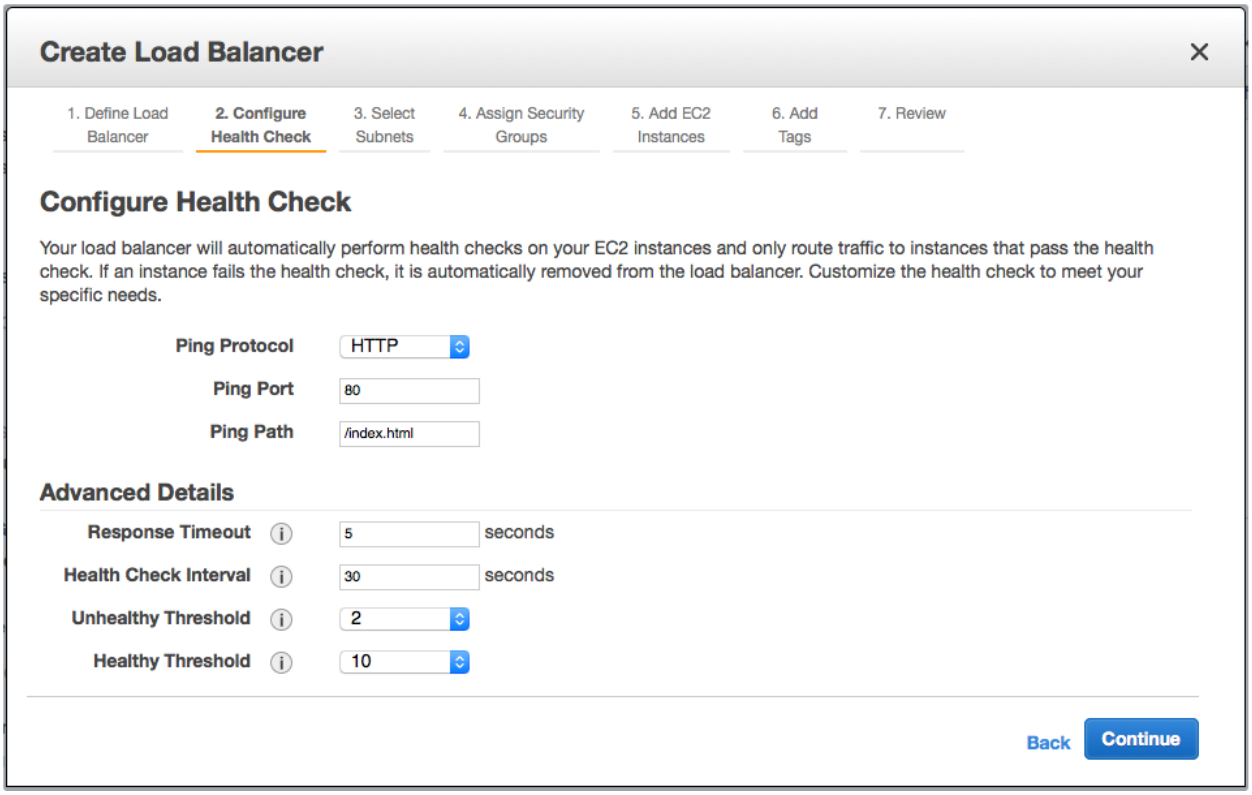
1. From the console dashboard, click **Load Balancers** link under "NETWORK & SECURITY" group.



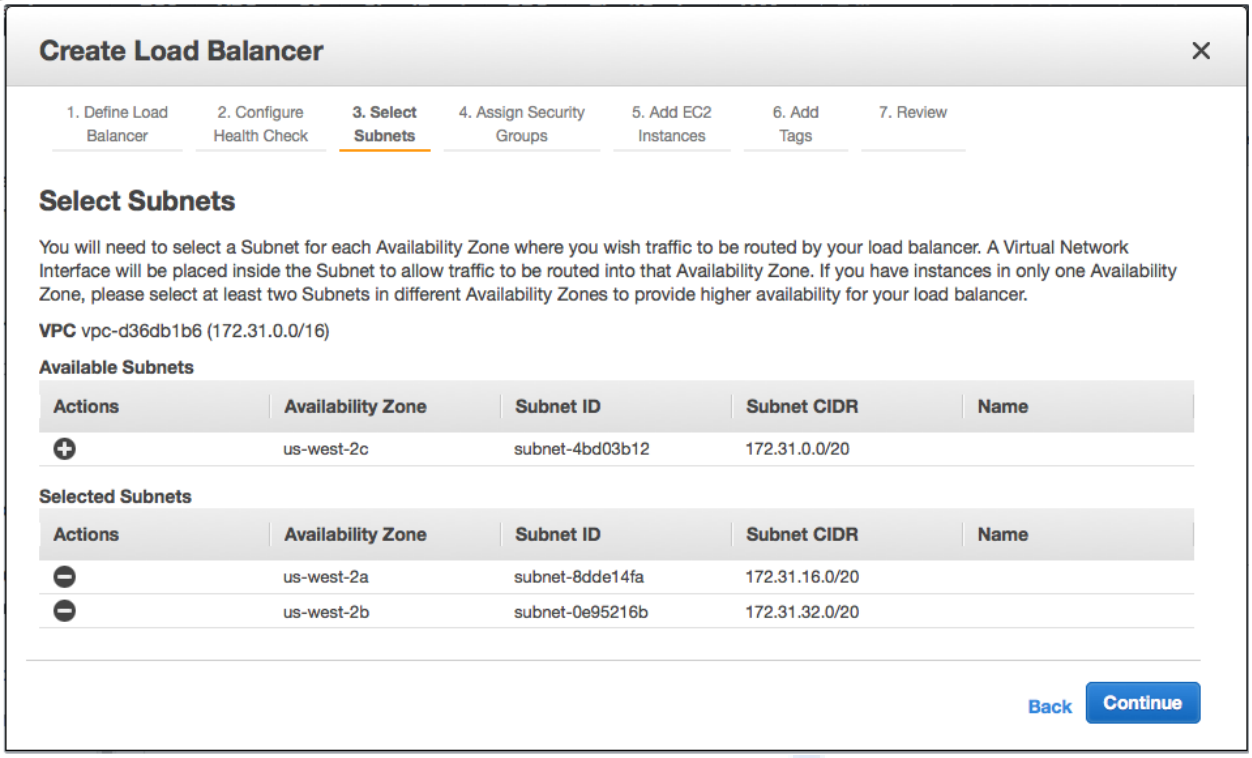
1. The list of all already created **Load Balancers** appears and you have to click **Create Load Balancer** blue button.
2. On the **Define Load Balancer** section, choose the load balancer name and **Enable advanced VPC configuration** and then click **Continue**.



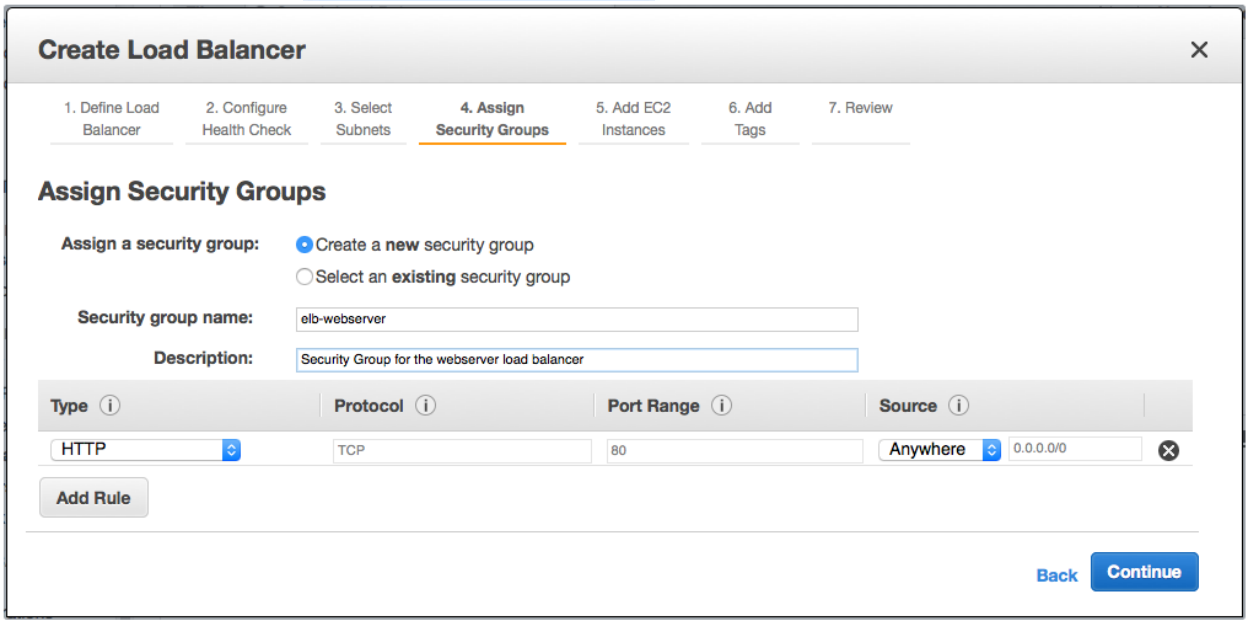
On the **Configure Health Check** section, replace the default value of **Ping Path** with a single forward slash ("/") and click



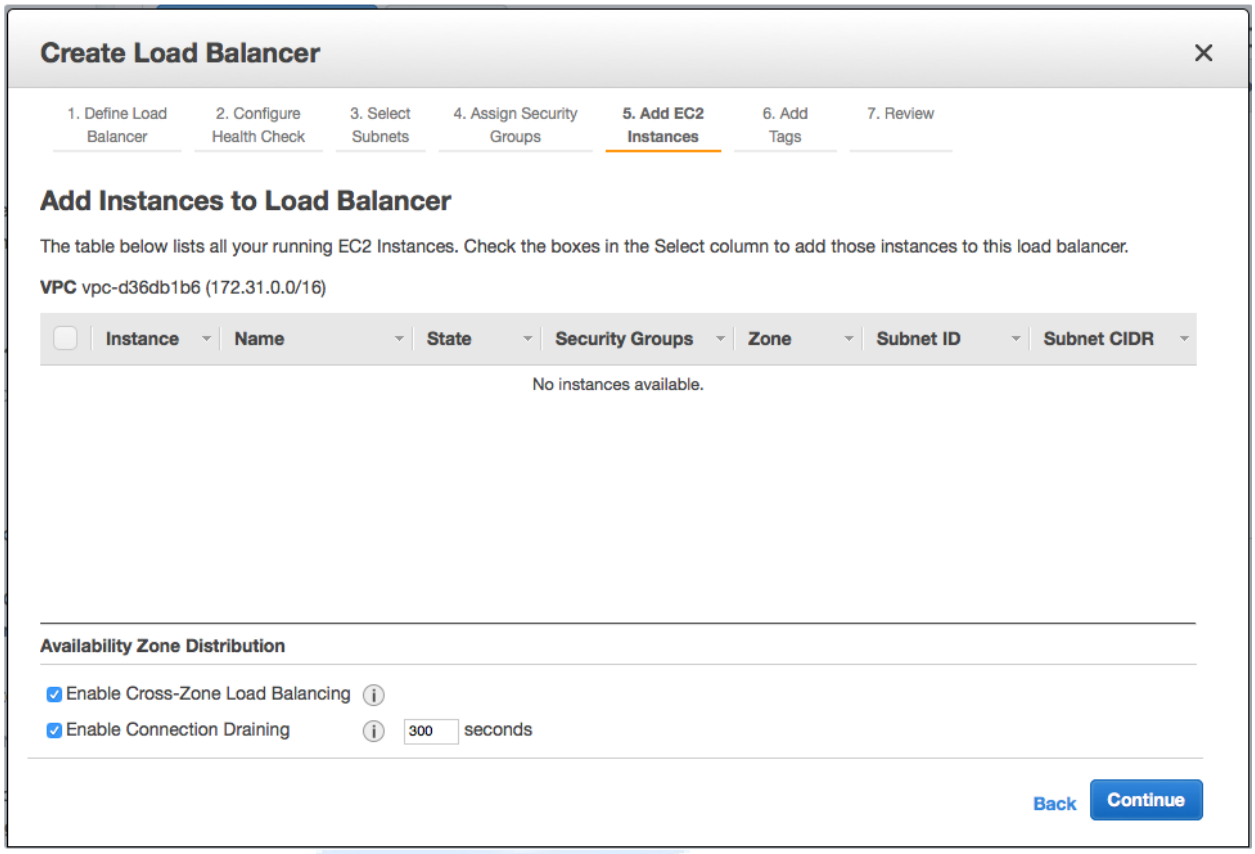
On the **Select Subnets** section, select all the subnets where the EC2 backend instances are.



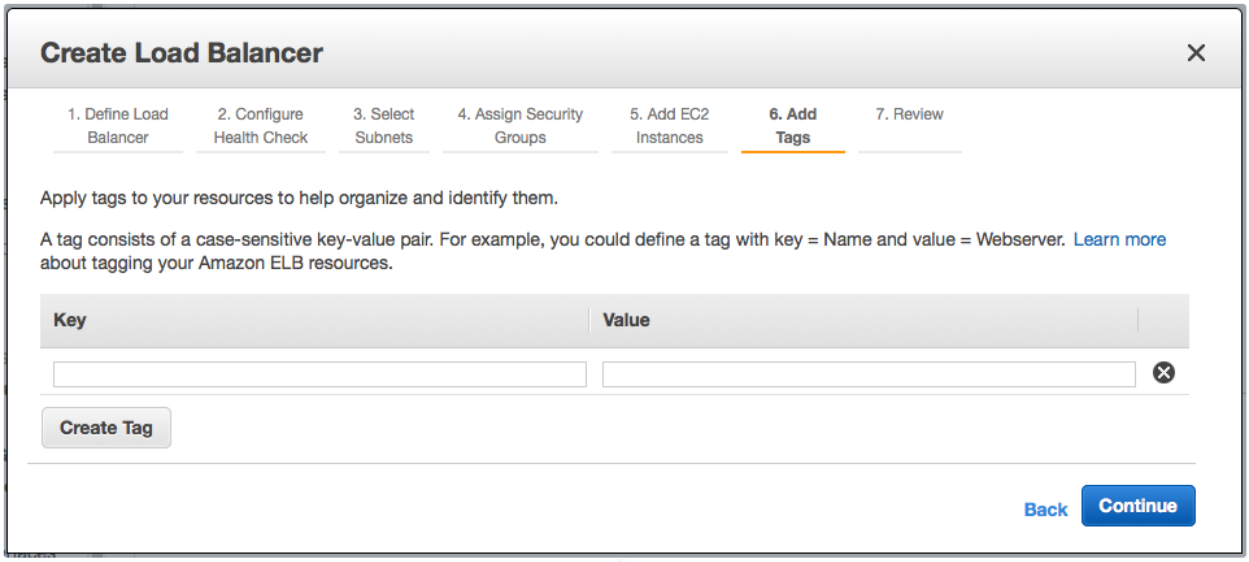
On the **Assign Security Groups** section, select **Create a new security group**, choose a name and a description and click



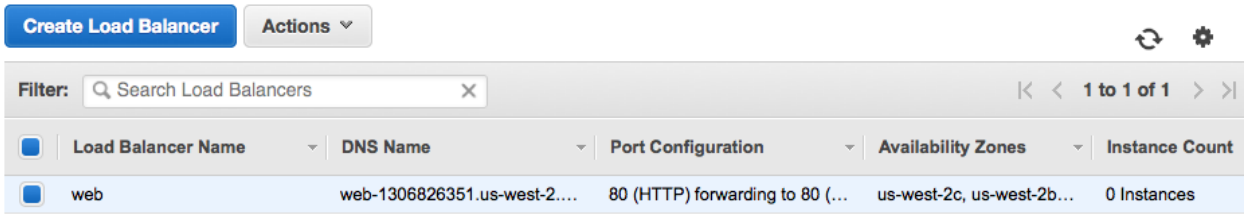
On the **Add EC2 Instances** section, select any EC2 instances that you want to balance (you can also add them later) and then click **Continue**.



On the **Add Tags** section, simply click **Continue** and then the **Create** button.



Now that you've created your load balancer, you're ready to verify its settings. The **Create Load Balancer** wizard displays the status of your newly created load balancer.
Click **Close** after confirming that your load balancer was successfully created.

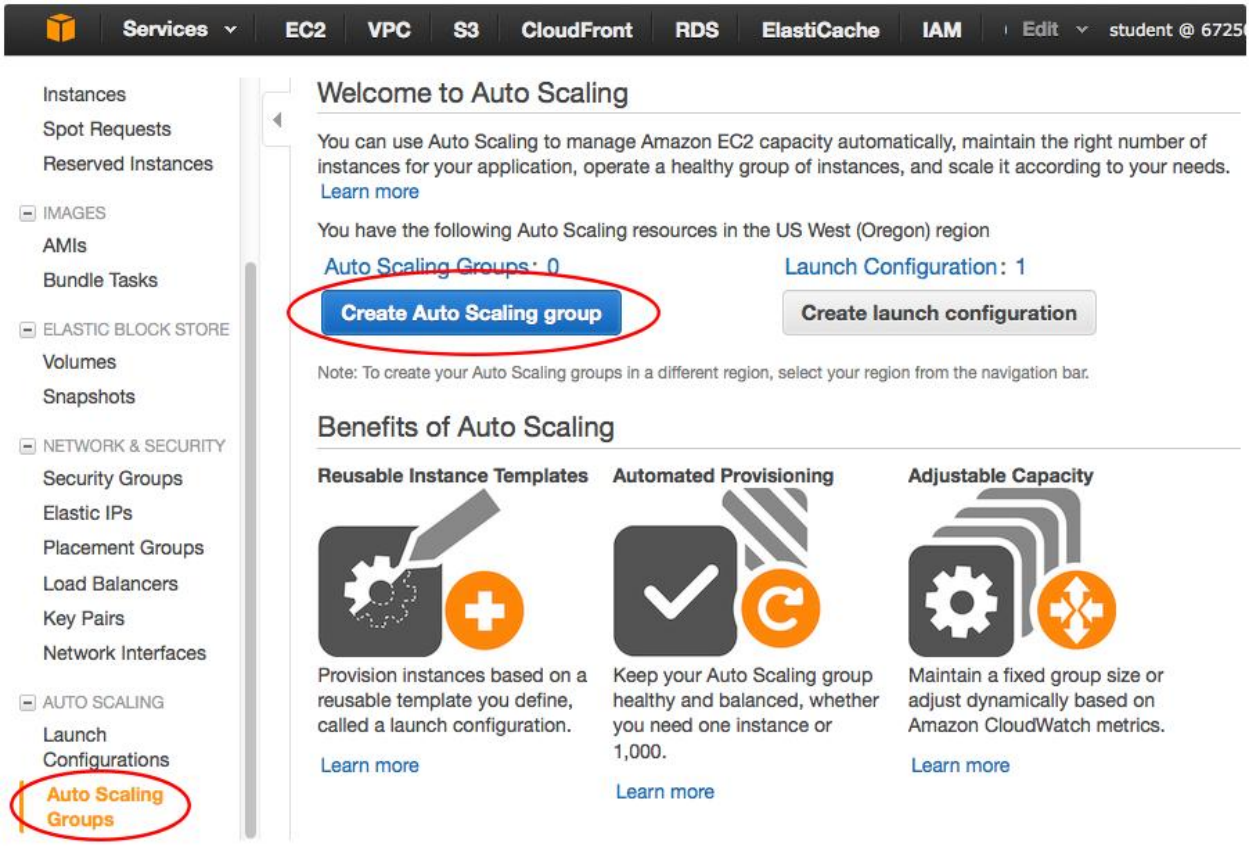


Create an Auto Scaling Group

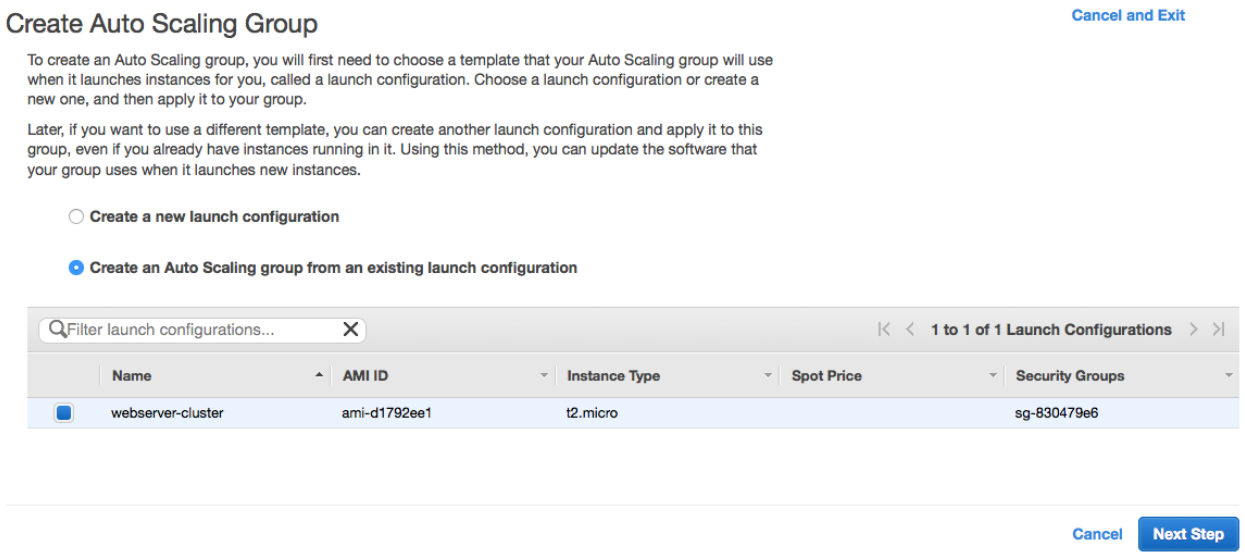
An Auto Scaling group is a representation of multiple Amazon EC2 instances that share similar characteristics, and that are treated as a logical grouping for the purposes of instance scaling and management. For example, if a single application operates across multiple instances, you might want to increase or decrease the number of instances in that group to improve the performance of the application. You can use the Auto Scaling group to automatically scale the number of instances or maintain a fixed number of instances. You create Auto Scaling groups by defining the minimum, maximum, and desired number of running EC2 instances the group must have at any given point of time.
An Auto Scaling group starts by launching the minimum number (or the desired number, if specified) of EC2 instances and then increases or decreases the number of running EC2 instances automatically according to the conditions that you define. Auto Scaling also maintains the current instance levels by conducting periodic health check on all the instances within the Auto Scaling group. If an EC2 instance within the Auto Scaling group becomes

unhealthy, Auto Scaling terminates the unhealthy instance and launches a new one to replace the unhealthy instance. This automatic scaling and maintenance of the instance levels in an Auto Scaling group is the core value of the Auto Scaling service.

To create the Auto Scaling group, click on the **Auto Scaling Groups** link in the Auto Scaling menu group and then click "Create Auto Scaling group" blue button.

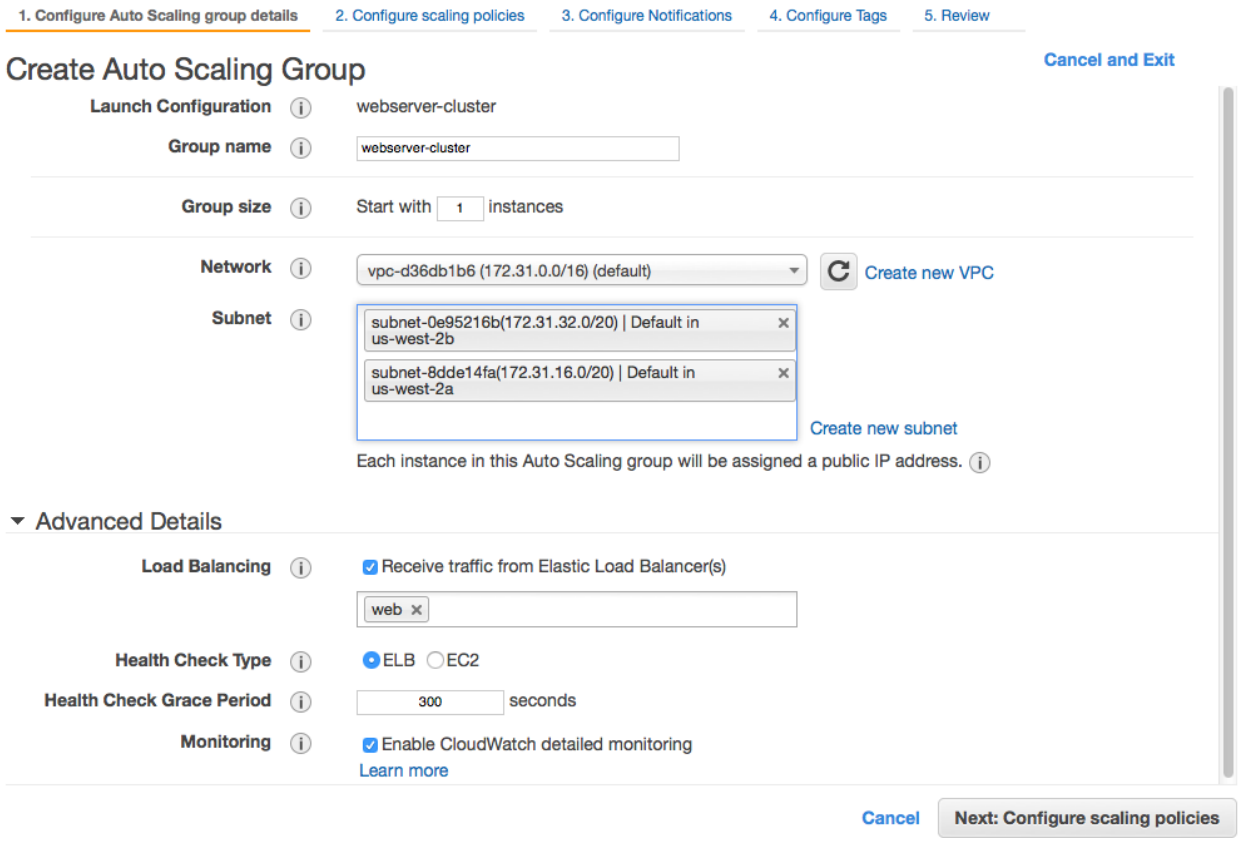


Select **Create an Auto Scaling group from an existing launch configuration** and then select the previously created launch configuration.



You need to choose the Auto Scaling group name, how many instances start, which VPS and subnet AWS should use during the deployment process.

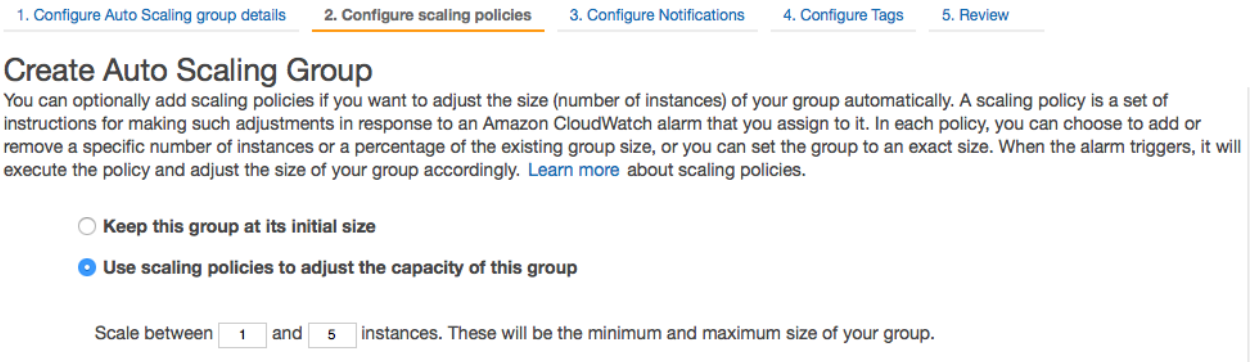
By opening the **Advanced Details** pane, you can select an **Elastic Load Balancer**, that you previously created, to automatically attach any created instance to it. You can decide to use the ELB health checks instead of the EC2 ones and enable the CloudWatch detailed monitoring.



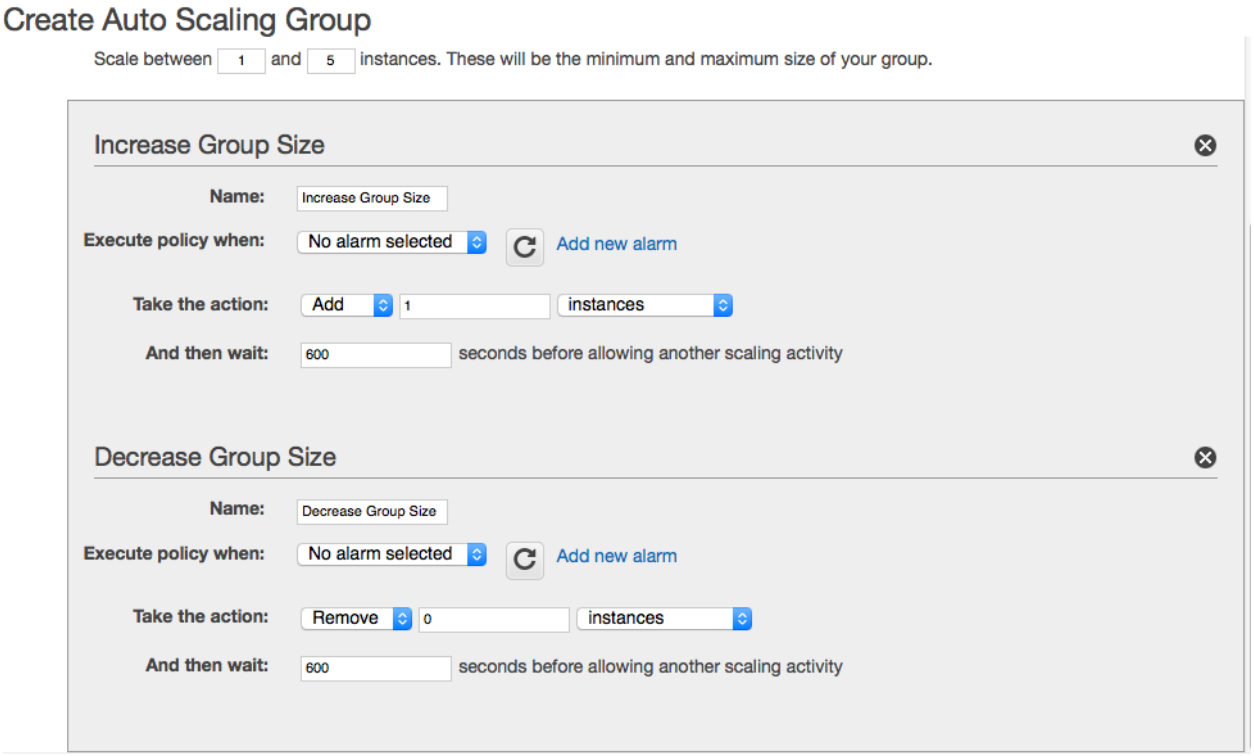
The second step is about configuring the scaling policies. You have to choose how much your infrastructure will be flexible and scale up or down.

Remember that each AWS Region has specific resource limits, so you cannot scale up to 1,000 instances for example. You should choose a minumim/maximum number of istances that isn't far away from the -/+20% of the average resource that you need now.

For this lab you can leave the default values: **Scale between 1 and 5 instances**.



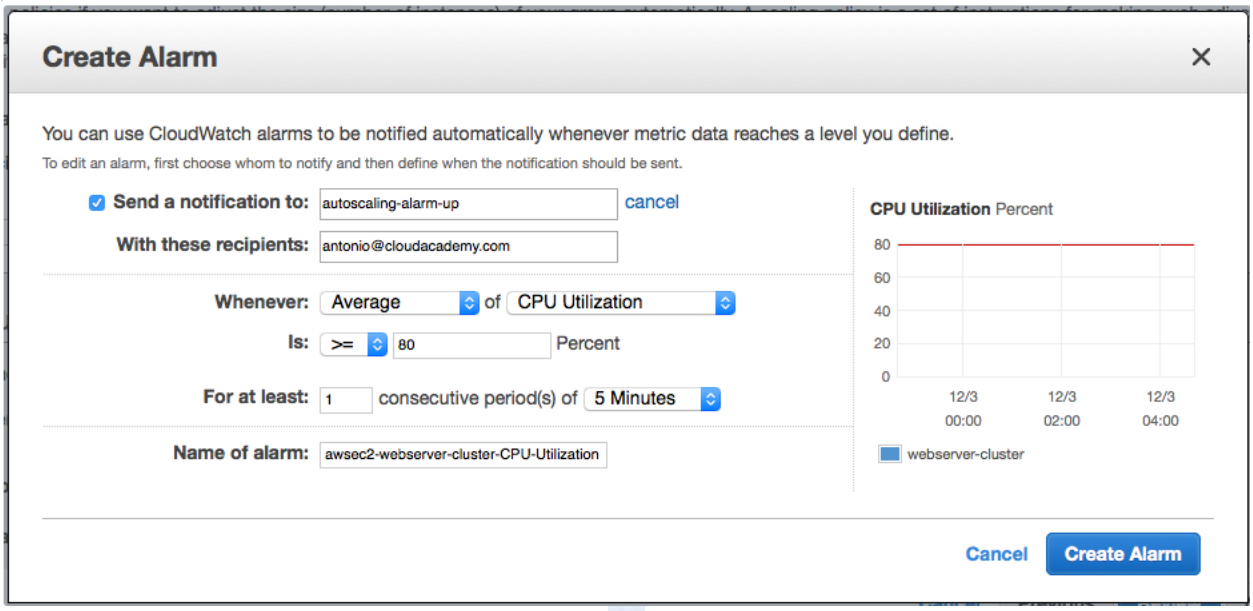
The Auto Scaling group policies allow you to automatically scale up or down your cluster. In order to do it, you have to create a CloudWatch Alarm a define which action should be taken if it is triggered.



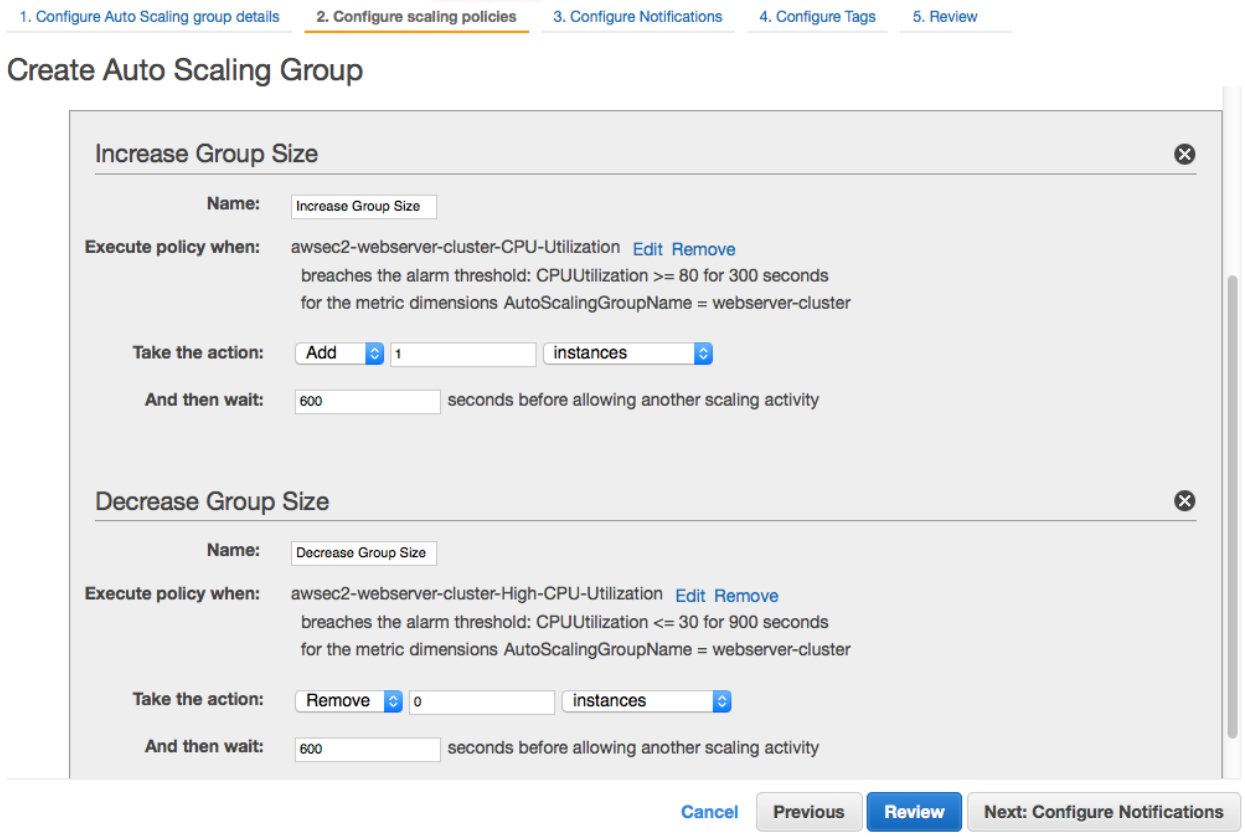
Adding an alarm is a quite easy task, click on **Add new Alarm** and fill the **Creale Alarm** form.

If you wan to receive a notification when the alarm is triggered, you need to setup a **SNS topic**: check the blue checkbox on top left, add the notification name and a mail recepient that will receive all the notifications.

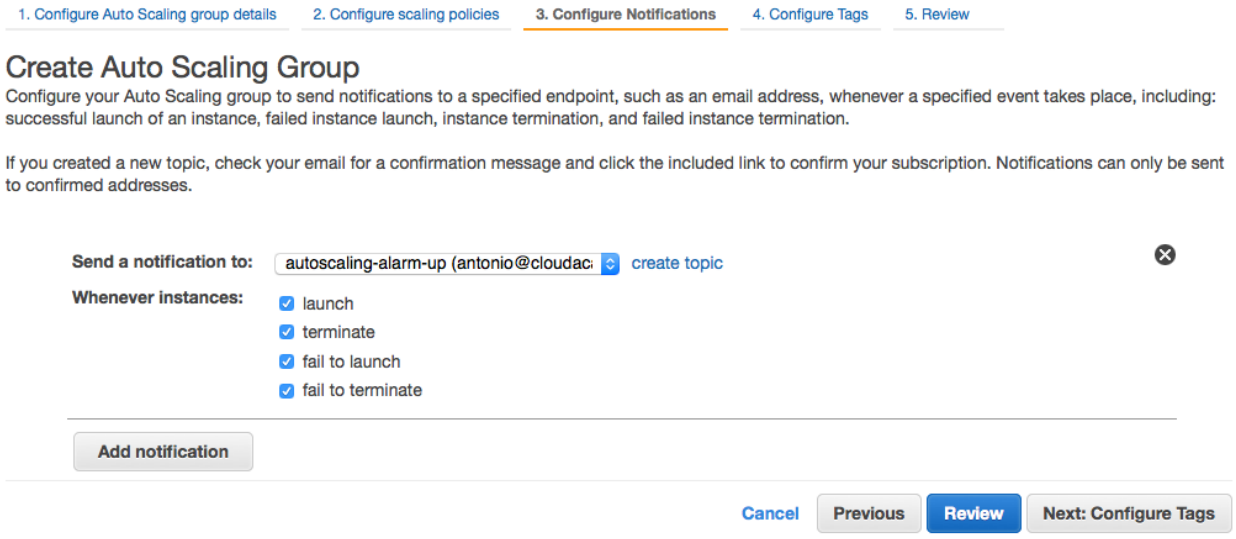
Select a metric (for example, CPUUtilization) and a constraint (Whenever **Average** Is >= **80** percent For at least **1**consecutive perdiod of **5 minutes**), choose the name of alarm and then click **Create Alarm.**



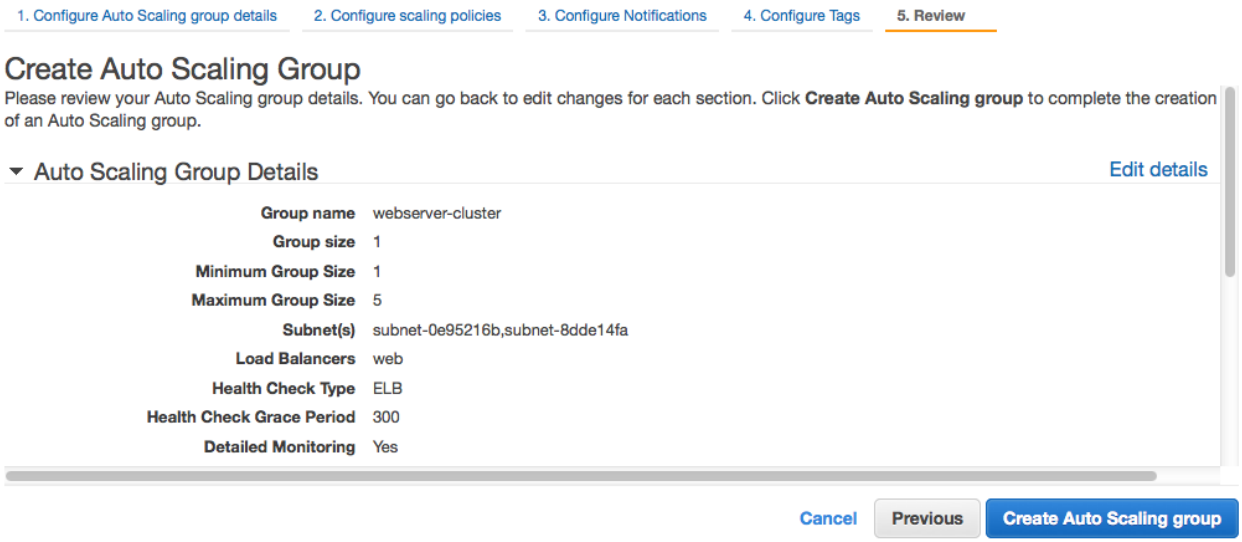
Create another alarm for the Decrease Group size and you are ready to go ahead.



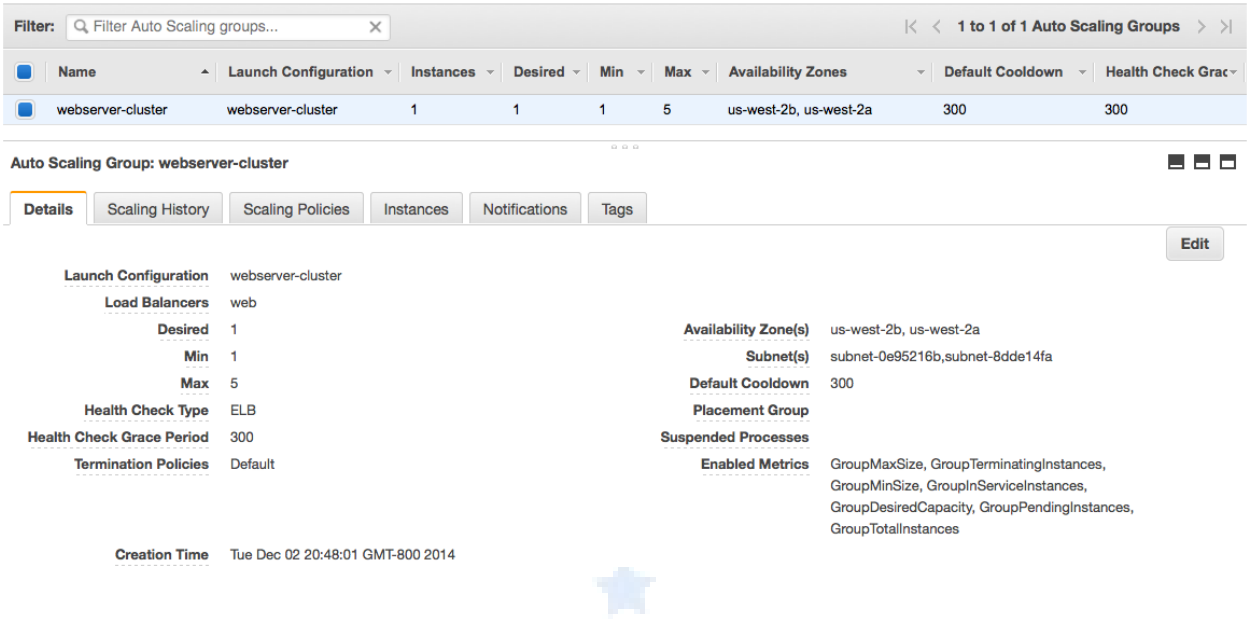
Configure Notifications allows you to be notified whenever an Auto Scaling Group instance is launched, terminated with or without success. You can use the same SNS topic previously created for the CloudWatch alarms. When you're done, click on the **Review** blue button.



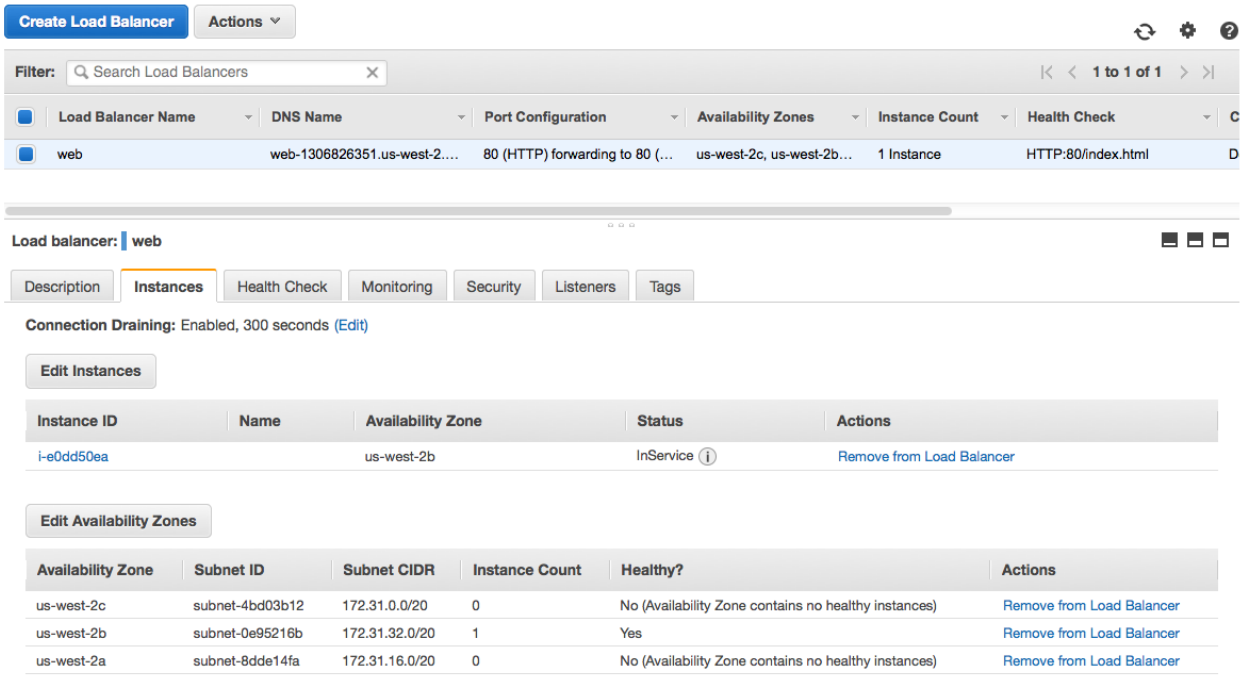
The **Review** tab allows you to check all the selected options and to start the creation of your cluster by clicking on Create Auto Scaling Cluster.



After some minutes your cluster will be deployed and all EC2 instances will be ready to be used.



By opening the **Load Balancers** section, selecting your previously created ELB and then opening the Instances tab, you can see the new Auto Scaling instances automatically added to its configuration.



SUVEN IT



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